

P11-B DE RUYTER, NetherlandsWMO: **062030**Lat: **52.360N**Lon: **3.342E**Elev: **79**StdP: **14.65**Time Zone: **1.00 (EUC)**Period: **09-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	29.8	32.2	18.5	14.0	32.2	22.3	16.8	34.1	47.3	46.6	43.0	46.3	20.9	60	0.789

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.7	70.5	64.8	68.1	63.4	66.5	62.6	65.9	69.0	64.5	67.2	63.4	66.0	14.3	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.5	91.1	68.0	63.2	87.1	66.5	62.2	83.9	65.4	30.7	69.1	29.7	67.4	28.9	66.1	71.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 39.8	36.0	32.4	DB	29.8	77.7	4.2	4.6	26.8	81.0	24.4	83.7	22.0	86.3	19.0	89.6	(4)
(5)			WB	26.3	67.9	4.7	2.4	22.9	69.6	20.2	70.9	17.5	72.3	14.1	74.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.2	42.9	41.5	43.9	47.8	52.2	57.2	61.8	63.6	61.2	56.8	50.3	46.2
	DBStd	8.49	4.65	4.95	4.09	4.20	4.12	3.58	3.20	2.62	3.04	3.67	4.26	4.71
	HDD50	939	220	238	191	90	23	1	0	0	0	2	45	129
	HDD65	4711	685	657	654	516	398	238	109	58	119	253	442	582
	CDD50	1735	0	0	2	23	90	216	367	422	336	214	53	12
	CDD65	35	0	0	0	0	0	3	11	15	6	0	0	0
	CDH74	32	0	0	0	0	0	3	21	4	4	0	0	0
	CDH80	4	0	0	0	0	0	0	4	0	0	0	0	0
Wind	WSAvg	16.7	19.9	19.6	16.8	15.0	14.1	13.6	13.1	14.5	15.4	17.9	18.9	21.5
Precipitation	PrecAvg	35.7	3.3	2.8	2.4	2.0	2.5	2.9	3.1	3.4	2.8	3.1	3.5	3.9
	PrecMax	45.0	7.2	6.3	5.6	4.5	5.2	5.2	5.9	7.9	7.8	6.1	5.7	8.5
	PrecMin	28.2	0.2	0.7	0.3	0.0	0.7	0.7	0.9	0.7	0.6	0.9	0.6	1.0
	PrecStd	4.4	1.5	1.3	1.2	1.1	1.0	1.2	1.3	1.8	1.5	1.1	1.3	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.1	52.0	54.7	62.6	65.4	70.9	76.9	73.6	71.9	66.3	60.6	55.6
		MCWB	49.7	47.8	49.1	53.0	56.8	63.3	68.2	65.4	66.4	61.7	57.8	52.8
	2%	DB	50.8	49.5	51.1	57.7	61.7	66.3	70.0	70.2	68.2	64.4	59.1	54.5
		MCWB	48.5	46.2	47.5	51.5	56.1	61.8	64.5	65.1	64.2	61.0	56.2	52.0
	5%	DB	49.8	48.2	49.6	54.5	59.0	62.8	67.3	67.9	66.4	63.0	57.9	53.4
		MCWB	47.6	45.4	46.7	50.3	55.0	59.5	63.2	63.4	62.4	59.6	54.8	51.0
	10%	DB	48.6	47.2	48.4	52.6	57.2	61.0	65.6	66.5	65.0	61.7	56.4	52.0
		MCWB	46.2	44.6	45.8	49.5	54.3	58.1	62.1	62.7	61.3	58.3	53.2	49.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.4	48.8	49.5	53.7	59.4	64.6	67.4	67.3	67.2	62.6	58.4	53.7
		MCDB	51.3	50.6	53.1	60.4	62.3	69.5	73.1	71.0	71.0	65.0	60.0	54.8
	2%	WB	49.3	46.8	48.3	52.0	56.8	62.1	65.4	65.6	65.4	61.3	56.7	52.6
		MCDB	50.4	48.5	50.7	55.9	59.2	65.4	68.9	68.5	67.8	64.0	58.7	54.0
	5%	WB	48.0	45.9	47.4	50.9	55.6	60.0	64.0	64.5	63.6	60.2	55.5	51.4
		MCDB	49.5	47.6	49.4	54.1	58.2	62.4	66.7	67.1	65.7	62.8	57.8	53.2
	10%	WB	46.7	45.0	46.3	49.7	54.4	58.4	62.7	63.5	62.3	58.9	53.7	49.7
		MCDB	48.3	46.9	48.1	52.4	57.0	60.4	64.8	66.2	64.6	61.3	56.3	51.7
Mean Daily Temperature Range	5% DB	MDBR	4.3	4.1	4.0	4.5	4.0	3.8	3.8	3.7	3.9	4.3	4.3	4.3
		MCDBR	5.1	4.4	5.1	8.0	7.5	7.3	6.9	5.6	4.5	4.9	5.1	4.8
		MCWBR	5.9	4.0	4.1	4.3	4.5	5.1	4.4	4.1	4.5	4.9	5.9	5.9
	5% WB	MCDBR	5.0	4.3	4.7	7.1	6.4	7.0	6.1	4.7	4.1	5.0	5.1	4.7
		MCWBR	6.1	4.1	4.1	4.1	4.4	5.5	4.5	4.5	4.6	5.3	6.0	6.1
Clear-Sky Solar Irradiance	taub	0.334	0.351	0.383	0.406	0.411	0.409	0.414	0.405	0.388	0.367	0.343	0.326	
	taud	2.354	2.372	2.305	2.279	2.332	2.352	2.348	2.386	2.422	2.425	2.381	2.342	
	Ebn at Noon	207	240	254	263	267	268	265	260	251	231	201	187	
	Edh at Noon	20	25	33	38	38	37	37	34	29	24	19	17	
All-Sky Solar Radiation	RadAvg	219	446	859	1429	1730	1840	1754	1482	1054	573	263	164	
	RadStd	18	79	88	125	108	97	184	105	71	54	25	18	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (72 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page